ELECTRONICS:

- BLDCs
- ESCs
- Propellers
- Radio Controllers
- GPS
- Telemetry
- Lidar
- Flight Controller
- Vtx and Fpv Camera
- Battery
- Power Distribution Board (PDB)

1) BLDCs -

BLDC = Brushless Direct Current motor
Stator: fixed coils | Rotor: permanent magnets
High efficiency and high power-to-weight

kV = RPM per Volt
Tells how fast a motor spins for each volt applied
Formula: RPM ≈ kV × Voltage
High kV - high speed, low torque (small props)
Low kV - low speed, high torque (large props)
Not kilovolt (very common confusion)

To change the direction of motors, swap any two wires !

2) ESCs -

Controls speed of a BLDC motor
Converts battery DC → 3-phase AC for the motor
Receives throttle commands from flight controller
Handles motor timing, braking, protection

PWM (Pulse Width Modulation)

- Analog-style control signal
- Speed set by pulse width (1000–2000 μs)
- Slower, noise-sensitive
- No feedback from ESC to controller
- Older but widely supported

DShot (Digital Shot)

- Digital ESC protocol (DShot150 / 300 / 600 / 1200)
- No calibration needed
- Error-checked, noise-immune
- Supports ESC telemetry (RPM, temp, current)

4in1 ESCs are not preferred in larger Quads - (Should only be equipped in fpv 5inch or smaller quads), as :

- Overheating risk in long flights
- Harder to repair during field operations
- Less suitable for high current, heavy prop

3) **Propellers** :

Diameter (inches) → size of prop
Bigger = more lift, more load
Pitch (inches) → how much air it pushes per rotation
Higher = more speed, more current
Written as Diameter × Pitch
Example: 10×4.5

Endurance → larger diameter, lower pitch
Speed / agility → smaller diameter, higher pitch

4) Radio Controllers :

CH1–4 → roll, pitch, throttle, yaw
Extra channels → flight modes, RTL, payload drop, etc.

Protocols :

ELRS (ExpressLRS) - UART COMMUNICATION - TX and RX

Modern Digital RC protocol (2.4 GHz / 900 MHz)
Very low latency
Very long range

PWM

One wire per channel, not compatible with px4
Pulse width ≈ 1000–2000 μs
Old, simple, bulky wiring

PPM
All channels on one wire, goes to rcin in pix4
Channels sent sequentially
Slower than SBUS
Rarely used now

SBUS
Digital protocol (by Futaba)
Up to 16 channels on one wire, goes to rcin in px4
Fast, reliable, low latency, but not as good as elrs

5) GPS:

Gives position (lat, lon), altitude, speed, time
Enables Loiter, RTL, Auto missions, geofencing
Often combined with a compass (magnetometer)
Communications via uart and can

6) TELEMETRY:

Wireless data link between drone and ground station
Sends live flight data + receives commands
Used with Mission Planner / QGroundControl
Upload missions, trigger RTL, modes
Communications done via tx, rx (UART)

7) LIDAR:

LiDAR = Light Detection and Ranging
Measures distance using laser pulses
Calculates distance by time-of-flight of light

Accurate altitude (especially below 20 m)
Obstacle avoidance, Terrain following, Precision landing
Mapping & 3D point clouds (advanced LiDARs)
Communications via UART or CAN

8) FLIGHT CONTROLLER:

Reads sensors → runs control algorithms → commands motors/servos
Keeps drone stable, controllable, and autonomous
Stabilizes drone (roll, pitch, yaw)

Converts RC / mission commands → motor outputs
Enables flight modes (Stabilize, Loiter, RTL, Auto)
Handles failsafes (RC loss, low battery)
Manages autonomous missions

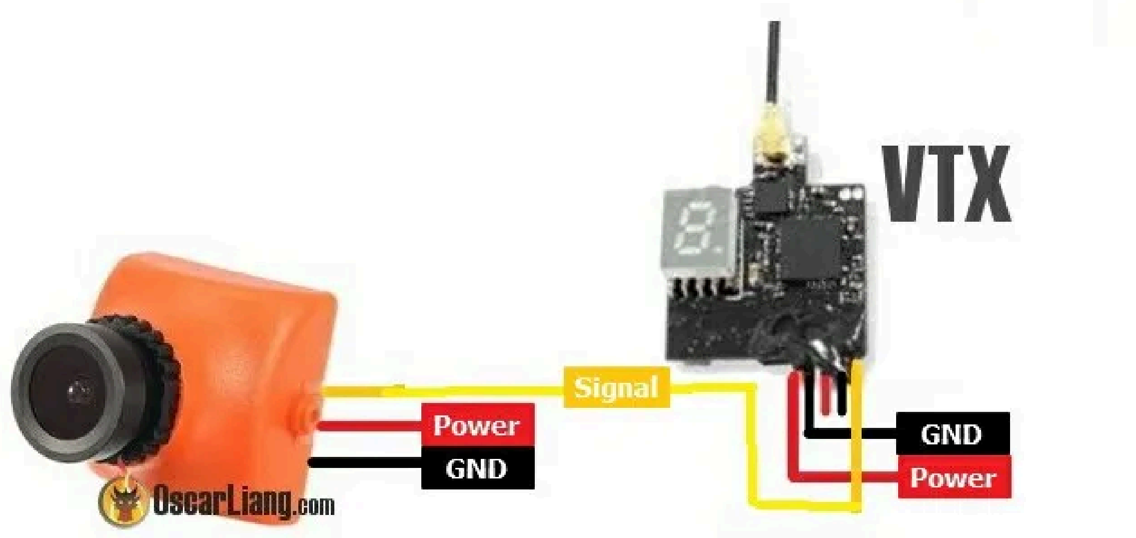
Why Pixhawk (specifically)

- Open-source (ArduPilot / PX4)
- Very reliable & widely tested
- Supports complex missions & autonomy
- Used in research, industry, competitions

IMU (Inertial Measurement Unit)

- Accelerometer → linear motion
- Gyroscope → angular rotation

9) VTx and FPV Camera :



Generally works in 5.8 Ghz frequency, is plug and play, just needs to set up the same freq. In video receiver and you are good to go

There is a search button on google, press it and it will automatically find that freq channel.

10. BATTERY:

Number of Cells (S Rating)

LiPo batteries are defined by the **number of cells connected in series**, written as **S**.

- **1S** = 3.7 V nominal
- **2S** = 7.4 V
- **3S** = 11.1 V
- **4S** = 14.8 V
- **6S** = 22.2 V

Important

- **Each LiPo cell has a nominal voltage of 3.7 V**
- Total battery voltage = `Number of cells × 3.7 V`
- Higher S → higher voltage → lower current for same power

Capacity (mAh)

Battery capacity is measured in **milliamp-hours (mAh)**.

- Example: **5000 mAh = 5 Ah**
- Indicates **how much charge the battery can store**
- Higher mAh:
    - Longer flight time
    - Higher weight

Capacity affects **endurance**, not thrust directly.

C Rating (Discharge Rating)

C rating defines **how fast current can be safely drawn** from the battery.

Formula

`Max Current (A) = Capacity (Ah) × C rating`

Example

- Battery: **5000 mAh (5 Ah), 30C**
- Max current = `5 × 30 = 150 A`

Notes

- Continuous C rating → safe continuous current
- Burst C rating → short-duration peaks
- Choose C rating based on **motor + ESC current demand**

Voltage Levels (Most Important Section)

Per-Cell Voltage Reference

Condition	Voltage (per cell)	Meaning
Fully Charged	**4.20 V**	Max safe voltage
Nominal	**3.70 V**	Normal operating
Storage Voltage	**3.80 V**	Long-term storage
Low Voltage	**3.50 V**	Land soon
Failsafe Trigger	**3.30 V**	RTL recommended
Critical	**3.20 V**	Battery damage
Absolute Minimum	**< 3.0 V**	Dangerous

Storage Voltage

LiPo batteries **must NOT be stored fully charged or fully discharged**.

- **Recommended storage voltage:**
  3.75 – 3.85 V per cell (typically 3.8 V)

Why?

- Prevents chemical degradation
- Reduces swelling
- Extends battery lifespan

Always use **storage mode** on LiPo chargers.

11. PDB:

Telemetry information sent to the Flight Controller and also powers the flight controller

A PDB **itself** usually does **not send telemetry**.
Telemetry comes from either:

- a **current & voltage sensor on the PDB**

What telemetry the flight controller receives:

- **Battery Voltage (V)**
- **Battery Current (A)**
- **Instant Power (W)** *(calculated internally)*
- **Consumed Capacity (mAh used)*

This information is sent to the **Pixhawk** via:

- **Analog voltage & current lines**
- Or via a **dedicated power module connector**

Pixhawk uses this data to:

- Trigger **low-battery warnings**
- Estimate **remaining flight time**
- Log **power consumption**
- Prevent deep battery discharge

Power supplied to ESCs

The **main job** of the PDB is power delivery to ESCs.

How ESCs are powered:

- PDB sends **raw battery voltage** directly to ESCs
  (e.g. 4S → ~16.8 V, 6S → ~25.2 V)

Important points:

- ESCs **do not receive regulated voltage**
- ESCs are designed to handle full battery voltage
- Each ESC draws only the current it needs

Flow:

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Battery → PDB → ESC → Motor
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